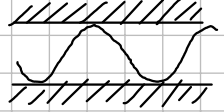


Lösungsvorschlag Probe-Prüfung Analysis

1. a) $\frac{6n^4 - 3n^2 + 7}{3n^4 - 2n} = \frac{\cancel{n^4}(6 - \frac{3}{n^2} + \frac{7}{n^4})}{\cancel{n^4}(3 - \frac{2}{n^3})} \xrightarrow{n \rightarrow \infty} \frac{6}{3} = \underline{2}$  Bei Folgen kein L'H

b) $\lim_{x \rightarrow 0} \frac{\sin x}{x} \stackrel{\frac{0}{0}}{=} \lim_{x \rightarrow 0} \frac{\cos x}{1} = \frac{\cos 0}{1} = \underline{1}$

c) $\lim_{x \rightarrow \infty} \frac{\cos x}{e^x} \stackrel{\text{"beschränkt" / "unendlich"}}{=} \underline{0}$ (da $\cos x \in [-1, 1]$ )

d) $\sum_{k=0}^{\infty} \frac{2^k}{k!} = e^2 \neq \cos(2)$ ODER: $\cos(2) < 0$ & $\sum_{k=0}^{\infty} \frac{2^k}{k!} = 1 + \frac{2}{1!} + \frac{2^2}{2!} + \dots > 1$

e) Wdh: Potenzreihe $\sum_{k=0}^{\infty} a_k (x - x_0)^k = \sum_{k=0}^{\infty} \frac{3^k}{(k+1)!} (x - 2)^k$ konvergiert für
 $x \in]x_0 - r, x_0 + r[$, wobei r der Konvergenzradius ist.

Berechne den Konvergenzradius r der Potenzreihe für die Koeffizienten $a_k = \frac{3^k}{(k+1)!}$:

$$\begin{aligned} r &= \lim_{k \rightarrow \infty} \left| \frac{a_k}{a_{k+1}} \right| = \lim_{k \rightarrow \infty} \frac{\frac{3^k}{(k+1)!}}{\frac{3^{k+1}}{(k+2)!}} = \lim_{k \rightarrow \infty} \frac{\frac{3^k}{(k+1)!}}{\frac{3(k+1)}{(k+2)!}} = \lim_{k \rightarrow \infty} \frac{k(k+2)}{k+1} \\ &= \lim_{k \rightarrow \infty} \frac{k \cdot (1 + \frac{2}{k})}{k \cdot (1 + \frac{1}{k})} = \infty \end{aligned}$$

$(k+2)! = (k+2)(k+1) \cdot k \cdot (k-1) \dots 3 \cdot 2 \cdot 1$
 $(k+1)! = (k+1) \cdot k \cdot (k-1) \dots 3 \cdot 2 \cdot 1$

Also konvergiert die Reihe für $x \in]\underset{\infty}{2-r}, \underset{\infty}{2+r}[= \mathbb{R}$.

2. a) Brauche Ableitungen von $h(x) = \ln x$:

$$h'(x) = \frac{1}{x} = x^{-1}, \quad h''(x) = -x^{-2} = -\frac{1}{x^2}, \quad h'''(x) = 2x^{-3} = \frac{2}{x^3}$$

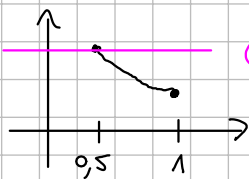
$$\begin{aligned} T_3(x) &= \sum_{k=0}^3 \frac{h^{(k)}(x_0)}{k!} (x-x_0)^k \\ &= h(1) + h'(1)(x-1) + \frac{h''(1)}{2}(x-1)^2 + \frac{h'''(1)}{6}(x-1)^3 \\ &= \underbrace{\ln(1)}_0 + \underbrace{\frac{1}{1}}_1 (x-1) + \underbrace{\frac{-1/1^2}{2}}_{-\frac{1}{2}} (x-1)^2 + \underbrace{\frac{2/1^3}{6}}_{\frac{1}{3}} (x-1)^3 \\ &= (x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 \end{aligned}$$

$$b) \ln(0,5) \approx T_3(0,5) = \underbrace{(0,5-1)}_{-0,5} - \frac{1}{2} \underbrace{(0,5-1)^2}_{0,25} + \frac{1}{3} \underbrace{(0,5-1)^3}_{-0,125} = -0,6$$

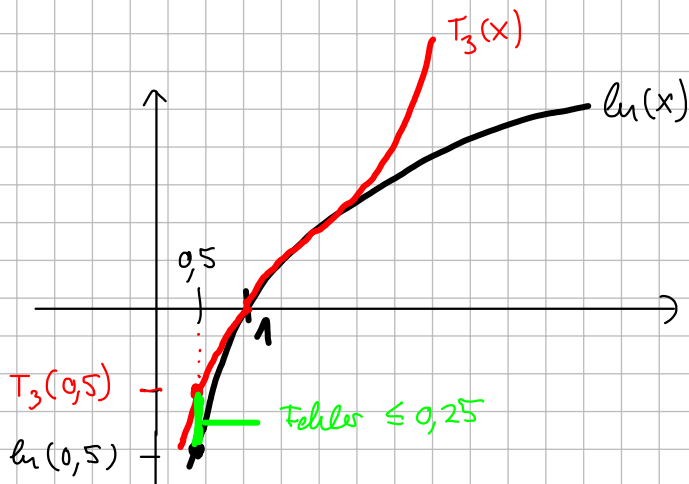
absoluter Fehler der Approximation ist:

$$|R_3(0,5)| \leq C \cdot \frac{|x-x_0|^4}{4!} = C \cdot \frac{|0,5-1|^4}{4!} = 96 \cdot \frac{0,0625}{24} = 0,25$$

zu C: $|h^{(4)}(x)| = |-6x^{-4}| = \left| -\frac{6}{x^4} \right| = \frac{6}{x^4}$ in $[0,5; 1]$ streng mo. ja:



$$C = \max_{x \in [0,5; 1]} \frac{6}{x^4} = \frac{6}{0,5^4} = 96$$



$$c) \quad T(x) = \sum_{k=1}^{\infty} \underbrace{\frac{(-1)^{k+1}}{k}}_{a_k} (x-1)^k = \frac{+1}{1} (x-1)^1 + \frac{-1}{2} (x-1)^2 + \frac{1}{3} (x-1)^3 + \dots$$

$$= (x-1) - \frac{1}{2} (x-1)^2 + \frac{1}{3} (x-1)^3 \mp \dots$$

Eine Taylorreihe ist eine Potenzreihe, somit berechnet man den Konvergenzradius

$$r = \lim_{k \rightarrow \infty} \left| \frac{a_k}{a_{k+1}} \right| = \lim_{k \rightarrow \infty} \left| \frac{\frac{(-1)^{k+1}}{k}}{\frac{(-1)^{k+2}}{k+1}} \right| = \lim_{k \rightarrow \infty} \underbrace{\frac{k+1}{k}}_{1 + \frac{1}{k} \downarrow 0} = 1.$$

$$\Rightarrow T(x) \text{ konvergiert f\"ur alle } x \in]x_0 - r, x_0 + r[=]1 - 1, 1 + 1[=]0, 2[.$$

3. $g(x) = x^2 e^x$

a) $\lim_{x \rightarrow -\infty} \underbrace{x^2}_{\downarrow \infty} \underbrace{e^x}_{\downarrow 0} = \lim_{x \rightarrow -\infty} \frac{x^2}{e^{-x}} \stackrel{\text{L'H}}{=} \lim_{x \rightarrow -\infty} \frac{2x}{-e^{-x}} \stackrel{\text{L'H}}{=} \lim_{x \rightarrow -\infty} \frac{2}{e^{-x}} = 0.$

$\lim_{x \rightarrow \infty} \underbrace{x^2}_{\downarrow \infty} \underbrace{e^x}_{\downarrow \infty} = \infty.$

b) NST: $\underbrace{x^2 e^x}_{>0} = 0 \Leftrightarrow x^2 = 0 \Leftrightarrow x = 0.$

lokale Extrema: $g'(x) = x^2 e^x + 2x e^x = (x^2 + 2x) \underbrace{e^x}_{>0} \stackrel{!}{=} 0 \Leftrightarrow$

$\Leftrightarrow \underbrace{x^2 + 2x}_{x(x+2)} = 0 \Leftrightarrow x = 0 \vee x = -2.$

$g''(x) = (x^2 + 2x) e^x + (2x + 2) e^x = (x^2 + 4x + 2) e^x$

$g''(0) = 2 > 0 \quad (\text{smiley}) \quad \text{an Stelle } x=0 \text{ Min.}$

$g''(-2) = ((-2)^2 + 4(-2) + 2) \frac{1}{e^2} = -2 \frac{1}{e^2} < 0 \quad (\text{sad}) \quad \text{an Stelle } x=-2 \text{ Max.}$

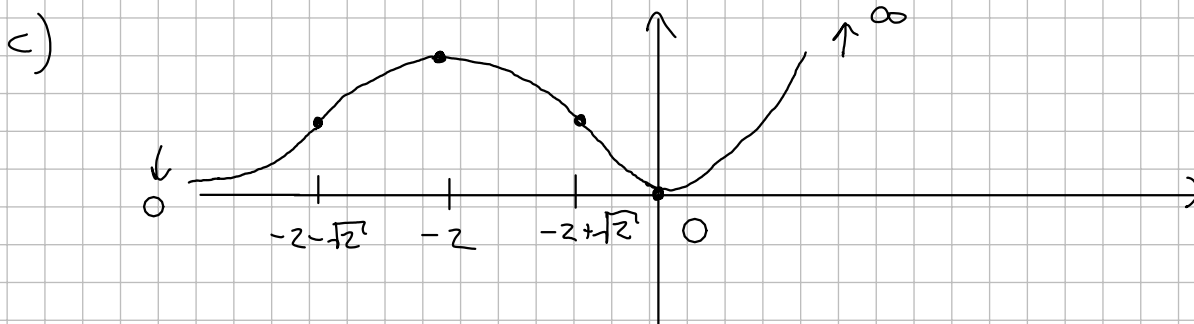
WP: $g''(x) = 0 \stackrel{e^x > 0}{\Leftrightarrow} x^2 + 4x + 2 = 0 \Leftrightarrow x_{1,2} = \frac{-4 \pm \sqrt{16 - 4 \cdot 2}}{2} = \frac{-4 \pm 2\sqrt{2}}{2} = -2 \pm \sqrt{2}$

$g'''(x) = (x^2 + 4x + 2) e^x + (2x + 4) e^x = (x^2 + 6x + 6) e^x$

$g'''(-2 + \sqrt{2}) = \underbrace{((-2 + \sqrt{2})^2 + 6(-2 + \sqrt{2}) + 6)}_{\approx 2,83} \underbrace{e^{-2 + \sqrt{2}}}_{>0} \neq 0$

$g'''(-2 - \sqrt{2}) = \underbrace{((-2 - \sqrt{2})^2 + 6(-2 - \sqrt{2}) + 6)}_{\approx -2,83} \underbrace{e^{-2 - \sqrt{2}}}_{>0} \neq 0$

$\Rightarrow$ zwei WP an den Stellen $x = -2 \pm \sqrt{2}$

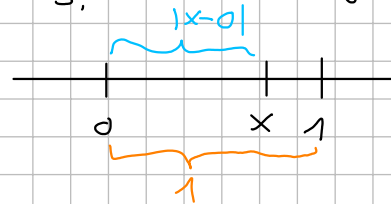


$$d) \quad T_2(x) = g(0) + g'(0) \cdot x + \frac{g''(0)}{2} x^2$$

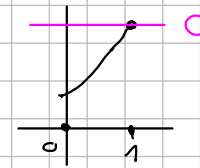
$$= 0 + 0 \cdot x + \frac{2}{2} x^2 = x^2$$

$$e) \quad \text{absoluter Fehler} = |R_2(x)| \leq C \cdot \frac{|x-0|^3}{3!} \leq 13 \cdot e \cdot \frac{1^3}{6} \approx \underline{5,88961...}$$

$$\bullet \quad |x-0| \leq 1 \quad \text{für } x \in [0,1]$$

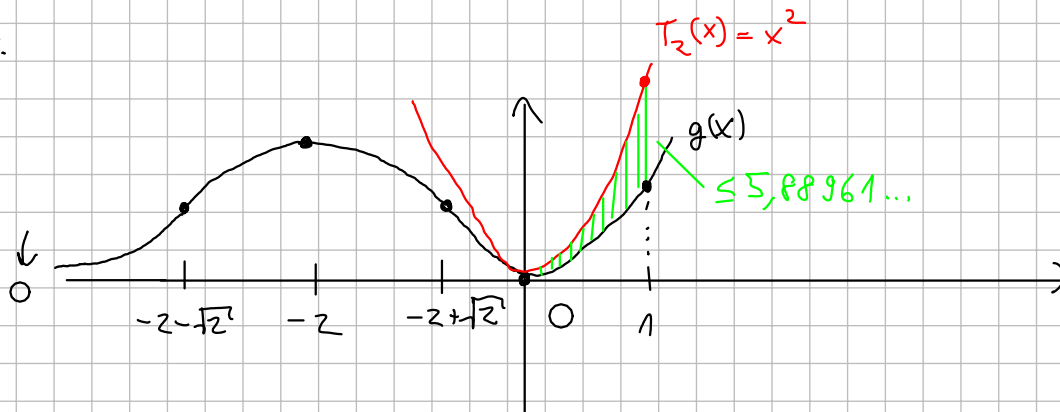


$$\bullet \quad C: \quad |g'''(x)| = \underbrace{|(x^2+6x+6)|}_{\substack{\text{str. mo. wa} \\ \text{in } [0,1]}} \underbrace{e^x}_{\substack{\text{str. mo. wa} \\ \text{in } [0,1]}} \quad , \text{ also}$$



$$\Rightarrow C = \max_{x \in [0,1]} |g'''(x)| = |g'''(1)| = \underbrace{(1^2+6 \cdot 1+6)}_{13} e^1 = \underline{13e \approx 35,33766...}$$

Also:



$$4. \quad f(x) = x^{-2} e^{x^2} = \frac{e^{x^2}}{x^2}.$$

a) $D = \mathbb{R} \setminus \{0\}$ wegen Nenner $x^2 \neq 0$.

b) $\lim_{x \rightarrow \pm\infty} \frac{e^{x^2}}{x^2} \stackrel{\text{L'H}}{=} \lim_{x \rightarrow \pm\infty} \frac{2x \cdot e^{x^2}}{2x} = \lim_{x \rightarrow \pm\infty} e^{x^2} = \infty$

$\lim_{x \rightarrow 0} \frac{e^{x^2}}{x^2} \stackrel{\text{"1/0"}}{=} +\infty.$

c) f ist stetig hebbar, falls $\lim_{x \rightarrow 0} \frac{e^{x^2}}{x^2} \in \mathbb{R}$ existieren würde. Also nein!

d) NST: $\frac{e^{x^2}}{x^2} = 0 \stackrel{x^2}{\Rightarrow} \underbrace{e^{x^2}}_{>0} = 0 \nrightarrow$ keine NST!

e) $\underline{\underline{f'(x) = \left(\frac{e^{x^2}}{x^2} \right)' = \frac{x^2 \cdot 2x \cdot e^{x^2} - e^{x^2} \cdot 2x}{(x^2)^2} = \frac{2x e^{x^2} (x^2 - 1)}{x^4} = \frac{2e^{x^2} (x^2 - 1)}{x^3}}}$

$\underline{\underline{f''(x) = \frac{x^3 \cdot [2 \cdot 2x \cdot e^{x^2} \cdot (x^2 - 1) + 2e^{x^2} \cdot (2x)] - 2e^{x^2} \cdot (x^2 - 1) \cdot 3x^2}{(x^3)^2} = \frac{2x^2 e^{x^2} \cdot \{x \cdot [2x(x^2 - 1) + 2x] - (x^2 - 1) \cdot 3\}}{x^6} = \frac{2e^{x^2} \cdot \{2x^4 - 3x^2 + 3\}}{x^4}}}$

f) lokale Extrema: $f'(x) = 0 \Leftrightarrow \underbrace{2e^{x^2}}_{>0} (x^2 - 1) = 0 \Leftrightarrow x^2 - 1 = 0$
 $\Leftrightarrow x^2 = 1 \Leftrightarrow x = \pm 1$

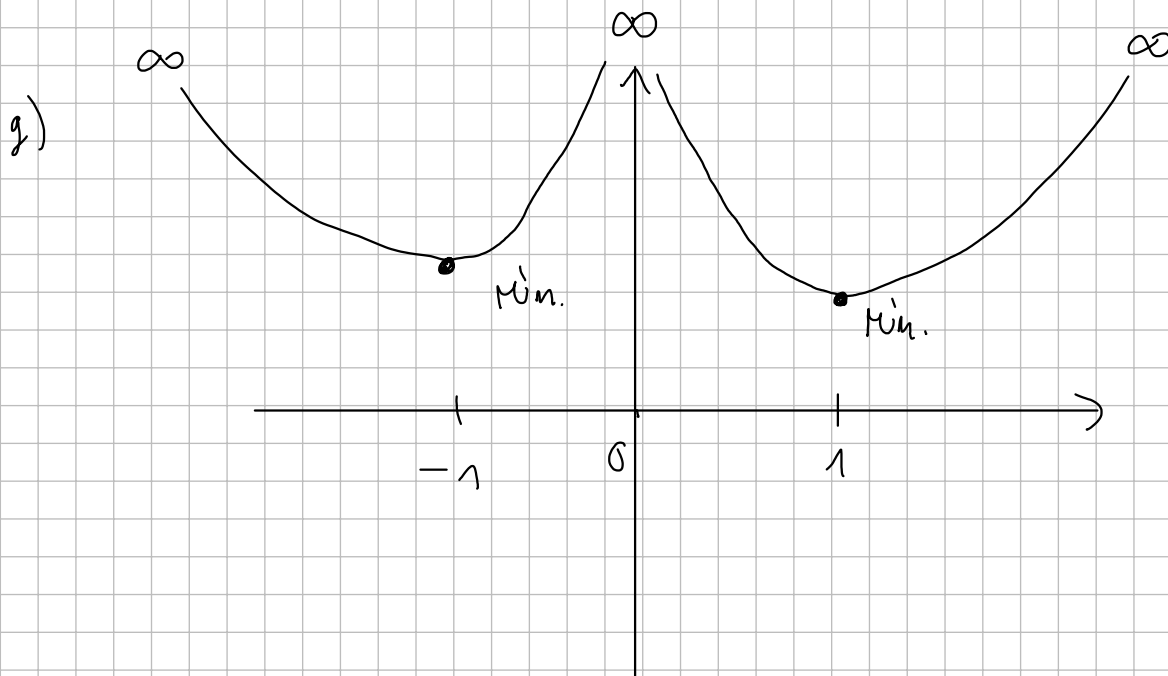
$$f''(\pm 1) = \frac{2e^{(\pm 1)^2} \cdot (2(\pm 1)^4 - 3(\pm 1)^2 + 3)}{(\pm 1)^4} = \frac{2e(2-3+3)}{1} = 4e > 0 \quad \text{😊}$$

$\Rightarrow$ an Stellen $x = \pm 1$ ist ein Minimum.

Wendepunkte: $f''(x) = 0 \Leftrightarrow \underbrace{2e^{x^2}}_{>0} (2x^4 - 3x^2 + 3) = 0 \Leftrightarrow 2x^4 - 3x^2 + 3 = 0$

Substitution $u = x^2$: $2u^2 - 3u + 3 = 0 \Rightarrow x_{1/2} = \frac{3 \pm \sqrt{9 - 4 \cdot 2 \cdot 3}}{4} < 0 \quad \swarrow$

$\Rightarrow$ keine Wendepunkte!



h)

$$T_2(x) = f(1) + f'(1) \cdot (x-1) + \frac{f''(1)}{2} (x-1)^2$$

$$= \frac{e^{1^2}}{1^2} + \frac{2e^{1^2}(1^2-1)}{1^3} (x-1) + \frac{\frac{2e^{1^2}(2 \cdot 1^4 - 3 \cdot 1^2 + 3)}{1^4}}{2} (x-1)^2$$

$$= e + 0 + 2 \cdot e (x-1)^2$$

$$= e + 2e(x-1)^2.$$

i)

$$f(1,01) \approx T_2(1,01) = e + 2e \cdot 0,01^2 \approx 2,718825484824737$$

$$f(1,01) = \frac{e^{1,01^2}}{1,01^2} \stackrel{TR}{\approx} 2,718823740130507$$

$$5. a) \int_0^2 z \sqrt{z^2+4} dz = \frac{1}{2} \int_0^2 2z \sqrt{z^2+4} dz = \frac{1}{2} \left[\frac{(z^2+4)^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^2$$

$$= \frac{1}{2} \left[\frac{(z^2+4)^{\frac{3}{2}}}{\frac{3}{2}} - \frac{(4)^{\frac{3}{2}}}{\frac{3}{2}} \right] = \frac{1}{3} \cdot (\sqrt{8}^3 - \underbrace{2^3}_8)$$

ODER: $\int_0^2 z \sqrt{z^2+4} dz = \int_{0^2+4}^{2^2+4} \frac{1}{2} \sqrt{u} du = \frac{1}{2} \left[\frac{u^{\frac{3}{2}}}{\frac{3}{2}} \right]_4^8 = \frac{1}{3} \cdot (\sqrt{8}^3 - \underbrace{2^3}_8)$

$u = z^2 + 4$
 $\frac{du}{dz} = 2z$
 $dz = \frac{du}{2z}$

$$b) \int \underset{f \cdot g'}{x e^x} dx = \underset{f \cdot g}{x e^x} - \int \underset{f' \cdot g}{1 \cdot e^x} dx = x e^x - e^x + C = e^x (x-1) + C$$

$$c) \int x e^{x^2} dx = \frac{1}{2} \int 2x e^{x^2} dx = \frac{1}{2} e^{x^2} + C$$

ODER: $\int x e^{x^2} dx = \int \frac{1}{2} e^u du = \frac{1}{2} e^u + C = \frac{1}{2} e^{x^2} + C.$

$u = x^2$
 $\frac{du}{dx} = 2x$
 $dx = \frac{du}{2x}$

$$d) \int \underset{f'}{\sin(x)} \underset{g}{(2-x)} dx = \underset{f \cdot g}{-\cos(x) \cdot (2-x)} - \int \underset{f \cdot g'}{(-\cos(x)) \cdot (-1)} dx = -\cos(x) \cdot (2-x) - \sin(x) + C$$

$$e) \int \frac{x+1}{2x^2+4x} dx = \frac{1}{4} \int \frac{4x+4}{2x^2+4x} dx = \frac{1}{4} \ln |2x^2+4x| + C$$

$(2x^2+4x)' = 4x+4$

f) nicht echt-rational $\rightarrow$ Polynomdivision:

$$\begin{array}{r} x^2 : (x^2 - 2x + 1) = 1 + \frac{2x-1}{x^2-2x+1} = 1 + \frac{2x-1}{(x-1)^2} \\ -(x^2 - 2x + 1) \\ \hline 2x - 1 \end{array}$$

$\uparrow$
doppelte NST des Nenners

$$\frac{2x-1}{(x-1)^2} \stackrel{!}{=} \frac{A}{x-1} + \frac{B}{(x-1)^2} = \frac{A(x-1) + B}{(x-1)^2} \Rightarrow \underline{2x-1 = A(x-1) + B}$$

$x = 1$ einsetzen liefert: $1 = B$

$x = 0$ einsetzen liefert: $-1 = A \cdot (-1) + 1 \Rightarrow A = 2$

$$\begin{aligned} \Rightarrow \int \frac{x^2}{x^2-2x+1} dx &= \int 1 + \frac{2}{x-1} + \frac{1}{(x-1)^2} dx = \\ &= x + 2 \ln|x-1| + \underbrace{\int 1 \cdot (x-1)^{-2} dx}_{\frac{(x-1)^{-2+1}}{-2+1} + C} = \\ &= \underline{x + 2 \ln|x-1| - \frac{1}{x-1} + C} \end{aligned}$$

g) Logarithmische Int. $(x^2+4x+5)' = 2x+4 \rightarrow$ Nein!

PBZ? NST Nenner: $x_{1,2} = \frac{-4 \pm \sqrt{16-4 \cdot 5}}{2}$ ^{<0} keine NST! PBZ: Hat schon Form $\frac{Ax+B}{x^2+4x+5}$.

$$\begin{aligned} \int \frac{2x+8}{x^2+4x+5} dx &\stackrel{\uparrow}{=} \int \frac{2x+4}{x^2+4x+5} + \frac{4}{x^2+4x+5} dx = \\ &\quad (x^2+4x+5)' = 2x+4 \quad \quad \quad \nwarrow \text{also quadratisch ergänzen } x^2+4x+5 = \underbrace{x^2+2 \cdot 2x+2^2+1}_{(x+2)^2} \\ &= \ln|x^2+4x+5| + 4 \cdot \underbrace{\int \frac{1}{(x+2)^2+1} dx}_{\arctan(x+2) + C} = \\ &= \underline{\ln|x^2+4x+5| + 4 \cdot \arctan(x+2) + C.} \quad \text{/oder: } u = x+2 \text{ substituieren} \end{aligned}$$